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LAB 7-Types of JOIN in SQL

- Inner Join

```
CREATE DATABASE IF NOT EXISTS store_db;
USE store_db;


DROP TABLE IF EXISTS Orders;
DROP TABLE IF EXISTS Users;


CREATE TABLE Users (
    user_id INT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(100)
);

CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    user_id INT,
    product_name VARCHAR(100),
    amount DECIMAL(10, 2),
    FOREIGN KEY (user_id) REFERENCES Users(user_id)
);


INSERT INTO Users (user_id, name, email) VALUES
(1, 'Sanjana', 'sanjana@gmail.com'),
(2, 'Sahasra', 'sahasra@gmail.com'),
```

```
(3, 'Ninada', 'ninada@gmail.com');

INSERT INTO Orders (order_id, user_id, product_name, amount) VALUES (101, 1,
'Smartphone', 699.99),
(102, 1, 'Headphones', 150.00),
(103, 2, 'Smartwatch', 249.50);


SELECT
    Users.user_id,


    Users.name,
    Users.email,
    Orders.order_id,
    Orders.product_name,
    Orders.amount
FROM Users
INNER JOIN Orders ON Users.user_id = Orders.user_id; ]
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

user_id	name	email	order_id	product_name	amount
1	Sanjana	sanjana@gmail.com	101	Smartphone	699.99
1	Sanjana	sanjana@gmail.com	102	Headphones	150.00
2	Sahasra	sahasra@gmail.com	103	Smartwatch	249.50

- Left Join

```
CREATE DATABASE IF NOT EXISTS store_db;
USE store_db;


DROP TABLE IF EXISTS Orders;
DROP TABLE IF EXISTS Users;


CREATE TABLE Users (
```

```
    user_id INT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(100)
);

CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    user_id INT,
    product_name VARCHAR(100),
    amount DECIMAL(10, 2),
    FOREIGN KEY (user_id) REFERENCES Users(user_id)
);


INSERT INTO Users (user_id, name, email) VALUES


(1, 'Sanjana', 'sanjana@gmail.com'),
(2, 'Sahasra', 'sahasra@gmail.com'),
(3, 'Ninada', 'ninada@gmail.com');

INSERT INTO Orders (order_id, user_id, product_name, amount) VALUES (101, 1,
'Smartphone', 699.99),
(102, 1, 'Headphones', 150.00),
(103, 2, 'Smartwatch', 249.50);


SELECT
    Users.user_id,
    Users.name,
    Users.email,
    Orders.order_id,
    Orders.product_name,
    Orders.amount
FROM Users
LEFT JOIN Orders ON Users.user_id = Orders.user_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

user_id	name	email	order_id	product_name	amount
1	Sanjana	sanjana@gmail.com	102	Headphones	150.00
1	Sanjana	sanjana@gmail.com	101	Smartphone	699.99
2	Sahasra	sahasra@gmail.com	103	Smartwatch	249.50
3	Ninada	ninada@gmail.com	NULL	NULL	NULL

- Outer Left Join

```
CREATE DATABASE IF NOT EXISTS store_db;
USE store_db;


DROP TABLE IF EXISTS Orders;
DROP TABLE IF EXISTS Users;


CREATE TABLE Users (
    user_id INT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(100)
);



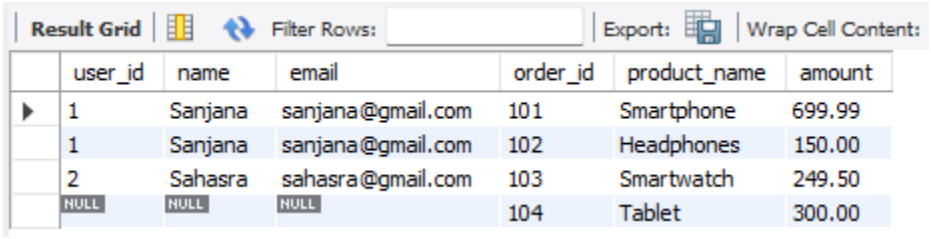
CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    user_id INT,
    product_name VARCHAR(100),
    amount DECIMAL(10, 2)
);


INSERT INTO Users (user_id, name, email) VALUES
(1, 'Sanjana', 'sanjana@gmail.com'),
(2, 'Sahasra', 'sahasra@gmail.com'),
(3, 'Ninada', 'ninada@gmail.com');


INSERT INTO Orders (order_id, user_id, product_name, amount) VALUES (101, 1,
'Smartphone', 699.99),
(102, 1, 'Headphones', 150.00),
(103, 2, 'Smartwatch', 249.50),
(104, 4, 'Tablet', 300.00);


SELECT
    Users.user_id,
```

```
    Users.name,
    Users.email,
    Orders.order_id,
    Orders.product_name,
    Orders.amount
FROM
    Users
        LEFT JOIN
    Orders ON Users.user_id = Orders.user_id
UNION SELECT
    Users.user_id,
    Users.name,
    Users.email,
    Orders.order_id,
    Orders.product_name,
    Orders.amount
FROM
    Users
        RIGHT JOIN
    Orders ON Users.user_id = Orders.user_id;
```



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

user_id	name	email	order_id	product_name	amount
1	Sanjana	sanjana@gmail.com	101	Smartphone	699.99
1	Sanjana	sanjana@gmail.com	102	Headphones	150.00
2	Sahasra	sahasra@gmail.com	103	Smartwatch	249.50
NULL	NULL	NULL	104	Tablet	300.00

- Cross Join

```
CREATE DATABASE IF NOT EXISTS store_db;
USE store_db;


DROP TABLE IF EXISTS Orders;
DROP TABLE IF EXISTS Users;
```

```
CREATE TABLE Users (
    user_id INT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(100)
);

CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    user_id INT,
    product_name VARCHAR(100),
    amount DECIMAL(10, 2)
);


INSERT INTO Users (user_id, name, email) VALUES
(1, 'Sanjana', 'sanjana@gmail.com'),
(2, 'Sahasra', 'sahasra@gmail.com'),
(3, 'Ninada', 'ninada@gmail.com');

INSERT INTO Orders (order_id, user_id, product_name, amount) VALUES



(101, 1, 'Smartphone', 699.99),
(102, 1, 'Headphones', 150.00),
(103, 2, 'Smartwatch', 249.50);

SELECT
    Users.user_id,
    Users.name,
    Orders.order_id,
    Orders.product_name,
    Orders.amount
FROM Users
CROSS JOIN Orders;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

user_id	name	email	order_id	product_name	amount
1	Sanjana	sanjana@gmail.com	102	Headphones	150.00
1	Sanjana	sanjana@gmail.com	101	Smartphone	699.99
2	Sahasra	sahasra@gmail.com	103	Smartwatch	249.50
3	Ninada	ninada@gmail.com	NULL	NULL	NULL
NULL	NULL	NULL	104	Tablet	300.00

```
CREATE DATABASE IF NOT EXISTS school_db;
USE school_db;


DROP TABLE IF EXISTS Teachers;


CREATE TABLE Teachers (
    teacher_id INT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(100),
    subject VARCHAR(50),
    salary DECIMAL(10, 2)
);


INSERT INTO Teachers (teacher_id, name, email, subject, salary) VALUES (1,
'Sanjana', 'sanjana@school.edu', 'Mathematics', 65000.00),
(2, 'Sahasra', 'sahasra@school.edu', 'Science', 72000.00),



(3, 'Ninada', 'ninada@school.edu', 'English', 58000.00);

-- 5. Subquery: Find teachers whose salary is greater than the average salary
SELECT
    teacher_id,
    name,
    subject,
    salary
FROM Teachers
WHERE salary > (
    SELECT AVG(salary)
    FROM Teachers
);
```